EXP.2. Numerical Simulation and Performance Analysis of Microwave Transmission Lines and Two-Port Networks Using Scattering Parameters.

A. Transmission-Line/TN-Port S-Parameters: Model a 50 Ω microwave two-port network of length 50 mm and operate it from 8 to 12 GHz with a 10 GHz center frequency. Set both ports to 50 Ω and obtain S11, S12, S21 and S22. At 10 GHz calculate return loss RL = $-20\log10|S11|$, insertion loss IL = $-20\log10|S21|$ and transmission efficiency $\eta = |S21|^2 \times 100\%$. Compare the simulated values with the ideal 50 Ω matched reference.

B. Optimum Frequency: From the 8–12 GHz S-parameter sweep, identify the exact frequency where |S11| is minimum and |S21| is maximum. Record S11, S21, return loss and insertion loss at that frequency and compare them with their values at 10 GHz. Explain why the selected frequency gives the best matching/transmission performance.

➔ **INPUT PARAMETER TABLE – VALUES TO BE ENTERED**

S.No.	Input Parameter	Symbol / Setting	Value to Enter	Unit
1	Operating frequency	f0	10	GHz
2	Reference impedance	Z0	50	Ω
3	Frequency sweep	fstart–fstop	8–12	GHz
4	Network length	L	50	mm
5	Port termination	ZL	50	Ω

➔ **OUTPUT PARAMETER TABLE – VALUES TO BE RECORDED**

S.No.	Output Parameter	Theoretical / Expected Value	Practical / Simulated Value	% Deviation
1	S11 at 10 GHz	≈ −20 dB or lower for good matching	Record simulation	% deviation
2	S21 at 10 GHz	≈ −1 to −3 dB depending on structure	Record simulation	% deviation
3	Return loss	$-20\log10\|S11\|$	Calculate	% deviation
4	Insertion loss	$-20\log10\|S21\|$	Calculate	% deviation
5	Best operating frequency	Frequency of minimum |S11| / maximum |S21|	Record	—

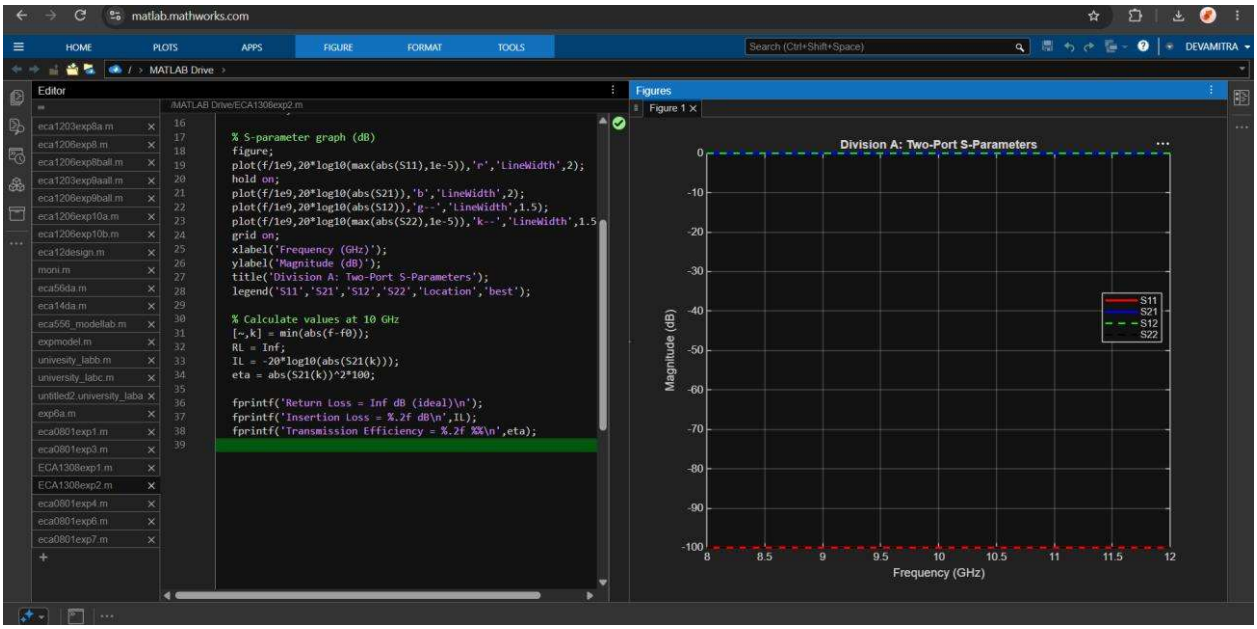
Aim: To examine the S-parameters of a 50-ohm microwave transmission line with a length of 50 mm over the 8 to 12 GHz frequency range. Return loss, insertion loss, and transmission efficiency are evaluated at 10 GHz, while the frequency sweep identifies the operating frequency offering the best reflection and transmission performance.

Software Required: MATLAB software.

Program A:

```
clc; clear; close all;
% Input parameters
f0 = 10e9;
f = linspace(8e9,12e9,501);
Z0 = 50; ZL = 50; L = 50e-3; c = 3e8;
beta = 2*pi*f/c;
% Ideal matched, lossless line
S11 = zeros(size(f));
S22 = zeros(size(f));
S21 = exp(-1j*beta*L);
S12 = S21;
% S-parameter graph (dB)
figure;
plot(f/1e9,20*log10(max(abs(S11),1e-5)),'r','LineWidth',2);
hold on;
plot(f/1e9,20*log10(abs(S21)),'b','LineWidth',2);
plot(f/1e9,20*log10(abs(S12)),'g--','LineWidth',1.5);
plot(f/1e9,20*log10(max(abs(S22),1e-5)),'k--','LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Division A: Two-Port S-Parameters');
legend('S11','S21','S12','S22','Location','best');
% Calculate values at 10 GHz
[~,k] = min(abs(f-f0));
RL = Inf;
IL = -20*log10(abs(S21(k)));
eta = abs(S21(k))^2*100;
fprintf('Return Loss = Inf dB (ideal)\n');
fprintf('Insertion Loss = %.2f dB\n',IL);
fprintf('Transmission Efficiency = %.2f %%\n',eta);
```

Output:

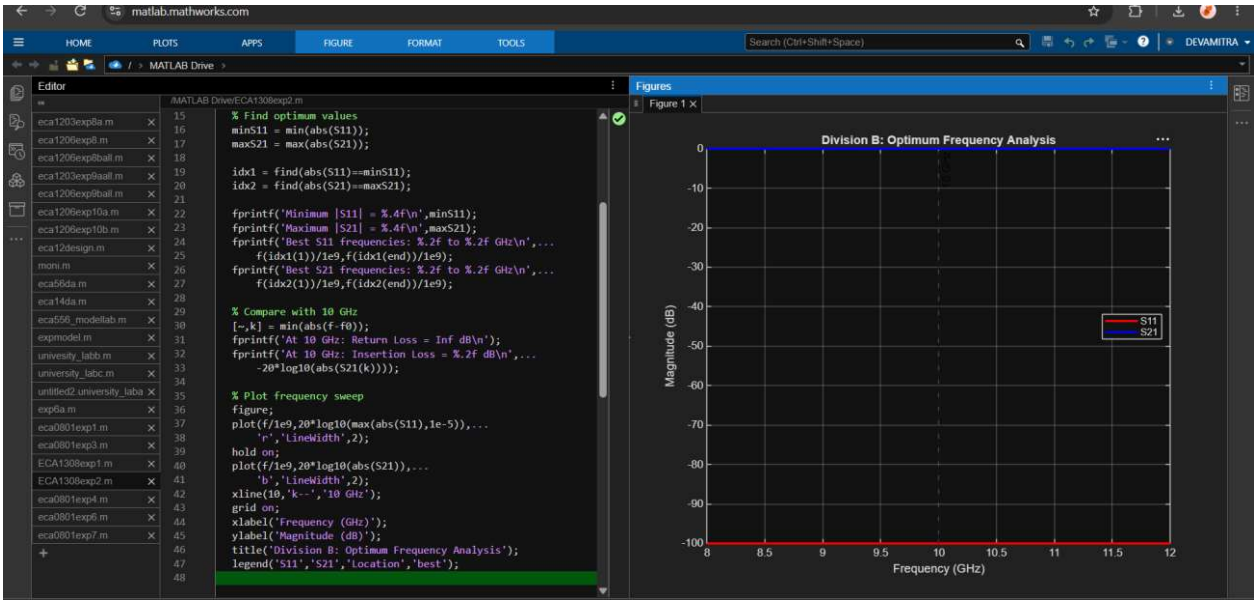


Program B:

```
clc; clear; close all;
% Input parameters
f0 = 10e9;
f = linspace(8e9,12e9,501);
Z0 = 50; ZL = 50; L = 50e-3; c = 3e8;
beta = 2*pi*f/c;
% Ideal matched, lossless line
S11 = zeros(size(f));
S21 = exp(-1j*beta*L);
% Find optimum values
minS11 = min(abs(S11));
maxS21 = max(abs(S21));
idx1 = find(abs(S11)==minS11);
idx2 = find(abs(S21)==maxS21);
fprintf('Minimum |S11| = %.4f\n',minS11);
fprintf('Maximum |S21| = %.4f\n',maxS21);
fprintf('Best S11 frequencies: %.2f to %.2f GHz\n',...
f(idx1(1))/1e9,f(idx1(end))/1e9);
fprintf('Best S21 frequencies: %.2f to %.2f GHz\n',...
f(idx2(1))/1e9,f(idx2(end))/1e9);
% Compare with 10 GHz
[~,k] = min(abs(f-f0));
fprintf('At 10 GHz: Return Loss = Inf dB\n');
fprintf('At 10 GHz: Insertion Loss = %.2f dB\n',...
-20*log10(abs(S21(k))));
% Plot frequency sweep
```

```
figure;
plot(f/1e9,20*log10(max(abs(S11),1e-5)),...
'r','LineWidth',2);
hold on;
plot(f/1e9,20*log10(abs(S21)),...
'b','LineWidth',2);
xline(10,'k--','10 GHz');
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Division B: Optimum Frequency Analysis');
legend('S11','S21','Location','best');
```

Output:



Result:

Division A: The S-parameters of the ideally matched 50-ohm two-port line were examined over 8 to 12 GHz. At 10 GHz, the ideal model produces zero reflection, infinite return loss, zero insertion loss, and 100 percent transmission efficiency.

Division B: The frequency sweep indicates zero reflection and maximum transmission across the full 8 to 12 GHz range for the ideal matched model. Thus, no single optimum frequency is identified, and performance at 10 GHz matches that at the other frequencies in the sweep.